SMART LIGHTING

by

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INTRODUCTION

Smart lighting is a key component of modern home automation.

- Home automation demand increasing since networking era
- Existing smart lighting systems are costly and complex
- Need for affordable and easy-to-install smart switch
- Smartphone control becoming standard feature

Limitations of conventional lighting systems

- Cannot control lights remotely
- High-voltage lights difficult to interface with controllers
- No systems that learn user lighting patterns
- Energy wasted when lights left ON in empty rooms

Problem Statement

The Problem Current lighting systems are inefficient. Conventional setups require manual effort, while existing smart solutions are often expensive, complicated, and lack the ability to learn user patterns. This leads to lights being left on unnecessarily, wasting energy.

The Solution There is a specific need for an **affordable, easy-to-install smart switch**. This device should offer remote control capabilities and intelligently automate lighting by detecting room occupancy and learning user habits.

Objectives of Project

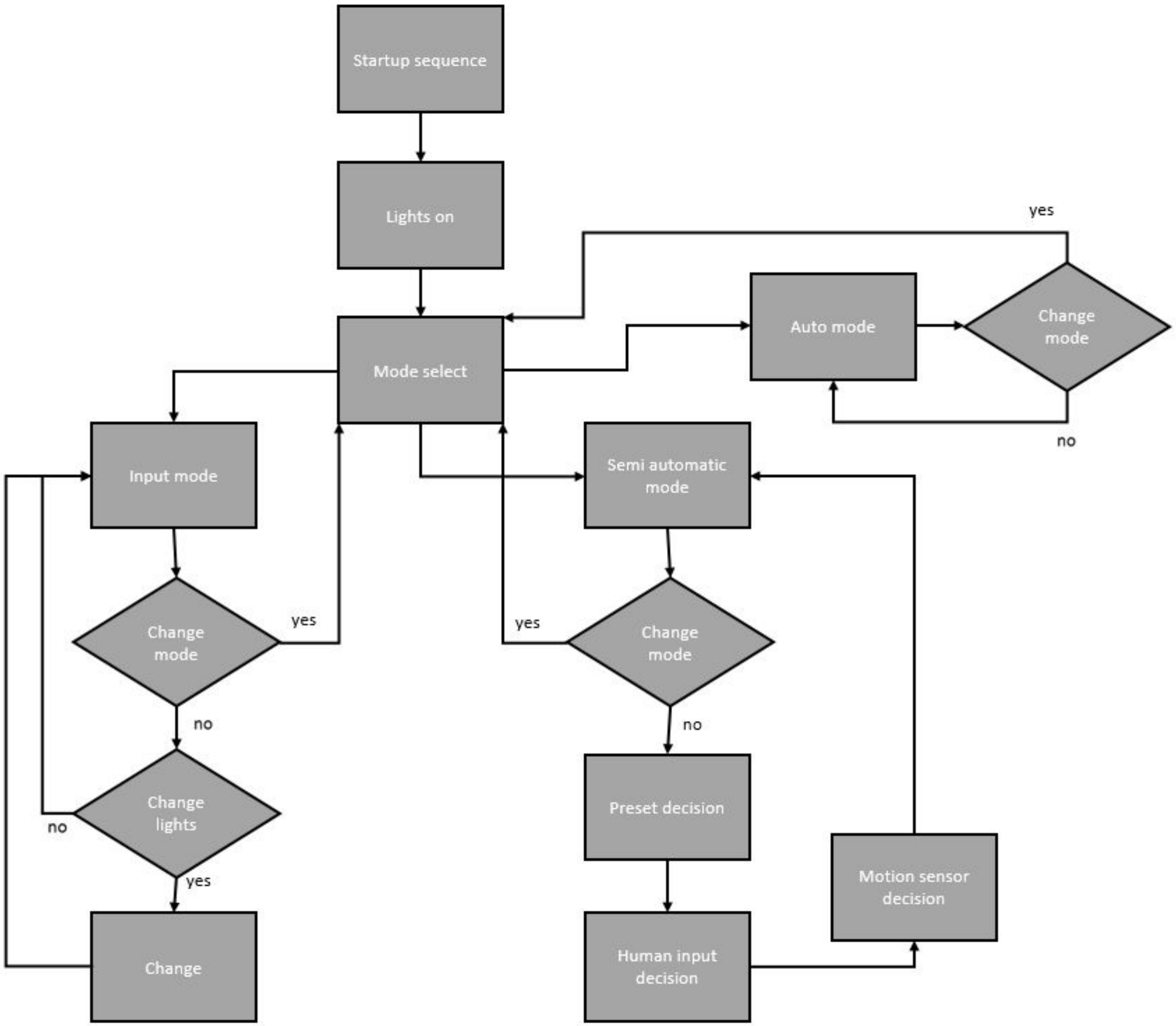
Goal: Develop a smart lighting switch prototype

- Replace conventional wall light switch
- Enable smartphone & manual control
- Detect motion and save energy
- Learn and predict lighting usage patterns

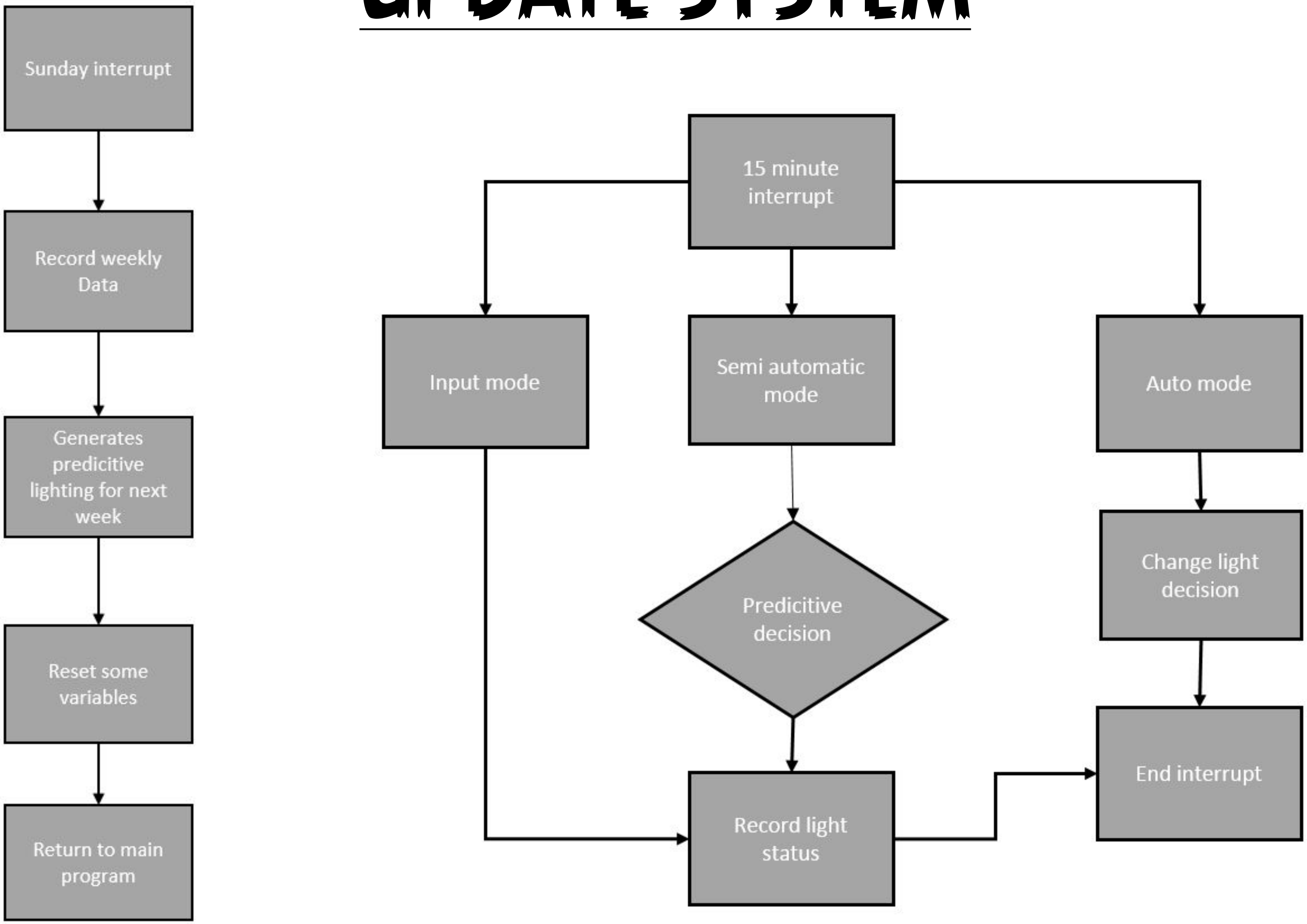
Literature Survey

Sr.No	Paper Name/title	Author Name/Year	Methodology	Future Scope/short comings	Results
1	**Smart Lighting System Using Arduino**	**Authors:**S. Jung, S. Lee, et al. **Year:** 2022	**Hardware:** Arduino Uno controller with Ultrasonic & IR sensors for presence detection. **Logic:** Uses a "Person-Counter" mechanism to track occupancy and toggle lights accordingly.	**Shortcomings:** Relies heavily on line-of-sight sensors; if the sensor is obstructed, the system fails to detect presence. **Future Scope:** Integration with dimming control based on sunlight intensity (LDR).	**Energy Savings:** Achieved a **50% reduction** in energy consumption (0.4 kWh to 0.2 kWh daily) by eliminating manual operation errors.
2	**Bluetooth Based Home Automation System**	**Authors:**Satish Kumar Singh et al. **Year:** 2019	**Comms:** Uses HC-05 Bluetooth module interfaced with Arduino to control home appliances via a custom Android App. **Control:** Simple ON/OFF switching via smartphone GUI.	**Shortcomings:** **Range Limitation:**Bluetooth restricts control to within ~10 meters (room-level only). **Security:** Basic Bluetooth protocols are susceptible to unauthorized pairing if not secured.	**Cost-Efficiency:** Proven to be the most **cost-effective** wireless solution compared to ZigBee or Wi-Fi for single-room automation, validating the low-budget approach.
3	**IoT-Based Smart Lighting System (ISLS)**	**Authors:**Liangliang Ma **Year:** 2024	**Connectivity:** Uses NodeMCU (Wi-Fi) to connect lighting systems to the Cloud (IoT). **Algorithm:** "Logic-based" method that uses decision trees to determine light levels based on time of day and occupancy.	**Shortcomings:** **Complexity:** Requires constant internet connection and cloud server setup, increasing initial cost and complexity. **Latency:** Dependent on network speed; internet outages disable remote control.	**User Comfort:** Demonstrated that automated "Adaptive Lighting" (changing brightness based on needs) improves user visual comfort and reduces eye strain significantly.

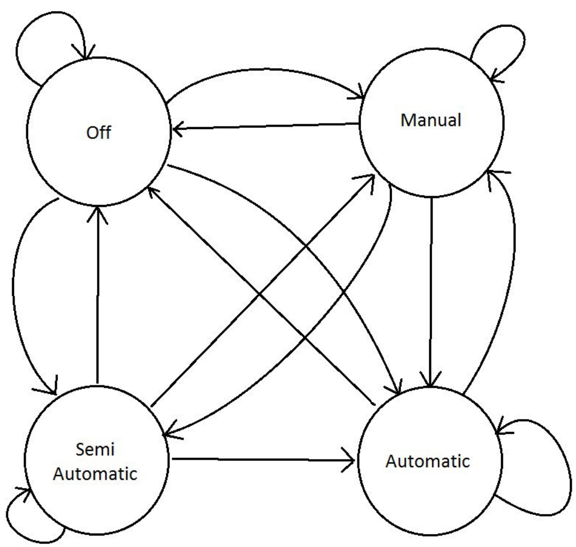
METHODOLOGY

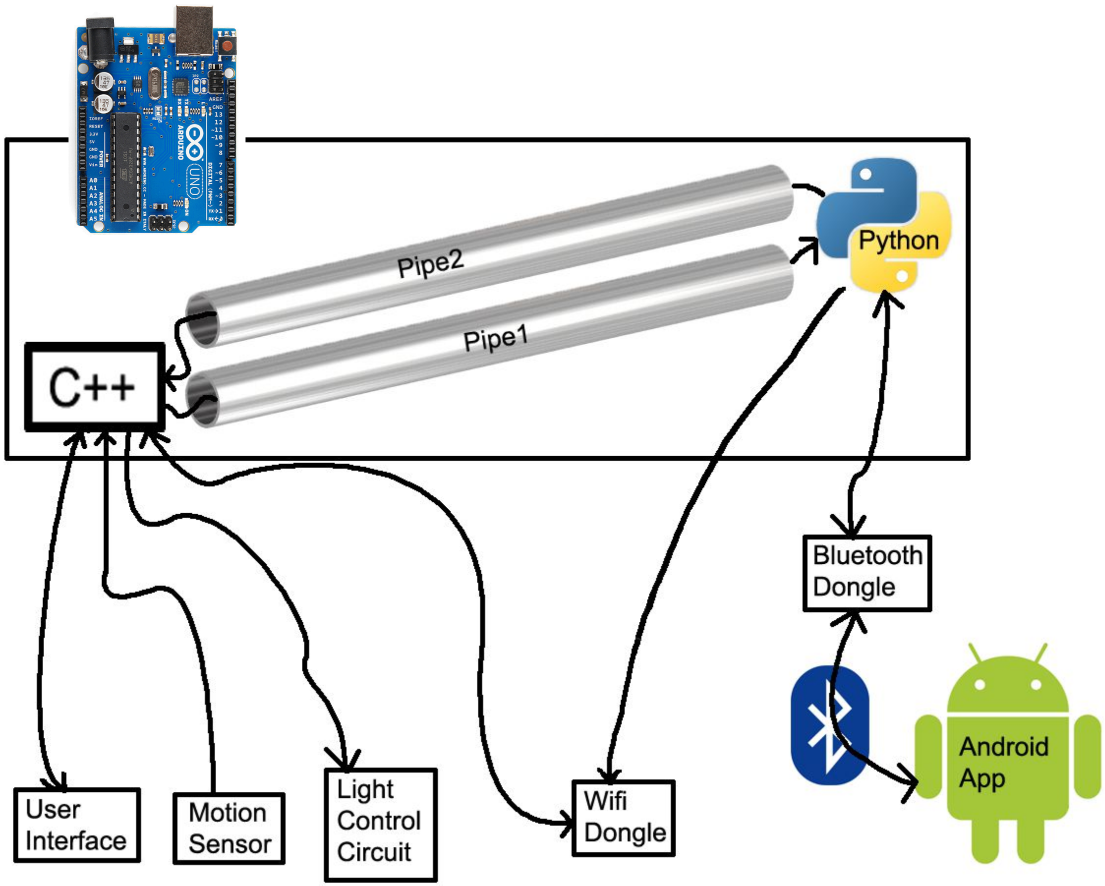


UPDATE SYSTEM



Methodology





Limitations

Limitations of the Proposed Smart Lighting System

- Prototype not integrated into a single compact wall-switch unit
- Prediction algorithm tested only in simulation (MATLAB), not real deployment
- Internet control removed due to development constraints
- Electrical safety certification not implemented in prototype
- Limited prediction accuracy (~60% for weekly events)
- Designed for single-room lighting control only

Application $ Future Scope

Applications of Smart Lighting System

- Smart home automation lighting control
- Energy-efficient lighting in homes and offices
- Security lighting (presence simulation when away)
- Automated room occupancy lighting
- Assistive lighting for elderly or disabled users
- Smart alarm system using light patterns

Future Scope and Enhancements

- Integration into standard wall switch form factor
- Cloud and mobile app remote control via Internet
- AI-based learning for higher prediction accuracy
- Multi-room and whole-home lighting networks
- Voice assistant integration (Alexa / Google Home)
- Energy monitoring and usage analytics

Conclusion

Smart switch replacing traditional switch

- Controlled via buttons or Android app
- Uses Arduino controller
- Motion sensor detects room occupancy
- Relay circuit controls lights

References

- **R. Heukels, “Predicting User Behavior Using Transition Probability,” University of Twente, 2015.**
  - • Provides the mathematical model for predicting user behavior from state transitions.
  - • Used as the basis for the lighting pattern recognition and prediction algorithm in this project.

- **Smart Lighting and Home Automation Literature**
  - • Industry and research sources describing smart lighting architectures and control methods.
  - • Provided background on motion sensing, remote control, and energy-efficient lighting automation concepts.